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SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - Configure a Wireless Network

Objectives

- Connect to a wireless router
- Configure the wireless router
- Connect a wired device to the wireless router
- Connect a wireless device to the wireless router
- Add an AP to the network to extend wireless coverage
- Update default router settings

Introduction

In this activity, you will configure a wireless router and an access point to accept wireless clients and route IP packets. Furthermore, you will also update some of the default settings.

Instructions

Part 1: Connect to a Wireless Router

Step 1: Connect Admin to WR.

- Connect **Admin** to **WR** using a straight-through Ethernet cable through the Ethernet ports. Select **Connections**, represented by a lightning bolt, from the bottom-left side of Packet Tracer. Click **Copper Straight-Through**, represented by a solid black line.
- When the cursor changes to connection mode, click **Admin** and choose **FastEthernet0**. Click **WR** and choose an available Ethernet port to connect the other end of the cable.

WR will act as a switch to the devices connected to the LAN and as a router to the internet. **Admin** is now connected to the LAN (**GigabitEthernet 1**). When Packet Tracer displays green triangles on both sides of the connection between **Admin** and **WR**, continue to the next step.

Note: If no green triangles are shown, make sure to enable **Show Link Lights** under **Options > Preferences**. You may also click **Fast Forward Time** just above the **Connections** selection box in the yellow bar.

Step 2: Configure Admin to use DHCP.

To reach the **WR** management page, **Admin** must communicate on the network. A wireless router usually includes a DHCP server, and the DHCP server is usually enabled by default on the LAN. **Admin** will receive IP address information from the DHCP server on **WR**.

- Click **Admin**, and select the **Desktop** tab.
- Click **IP Configuration** and select **DHCP**.

What is the IP address of the computer?

192.168.0.100

What is the subnet mask of the computer?

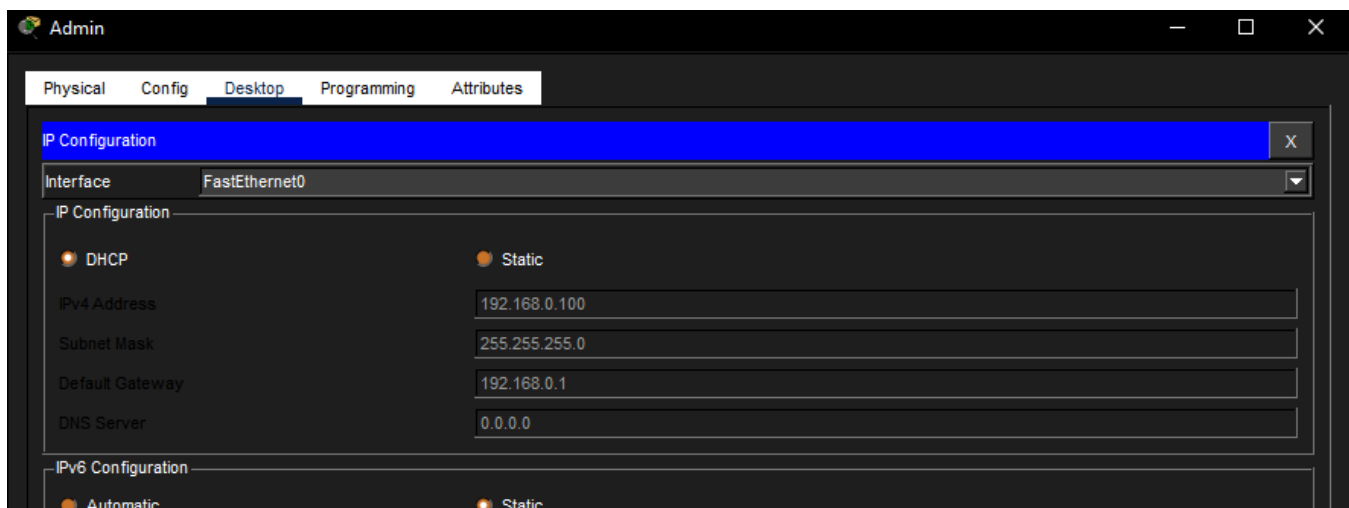
255.255.255.0

What is the default gateway of the computer?

192.168.0.1

- Close the **IP Configuration** window.

Note: Values can vary within the network range due to normal DHCP operation.



Step 3: Connect to the WR Web Interface.

- In the **Desktop** tab on **Admin**, choose **Web Browser**.
- Enter **192.168.0.1** in the URL field to open the web configuration page of the wireless router.
- Use **admin** for both the username and password.
- Under the Network Setup heading on the **Basic Setup** page, notice the IP address range for the DHCP server.

Is the IP address for **Admin** within this range? Is it expected? Explain your answer.

Yes, the IP address for the Admin PC is within the designated range, which is entirely expected. The Admin computer is explicitly set up to request its network configuration automatically via DHCP. By

default, wireless routers have an integrated DHCP server enabled to manage the Local Area Network (LAN). Due to this configuration, the router automatically assigns available IP addresses, subnet masks, and default gateways from its predefined pool to any connected client devices. Consequently, when the Admin PC connects and requests network details, receiving an IP address directly from that default range is the intended and standard behavior of the network.

The screenshot shows the Admin interface for a Wireless-N Broadband Router. The 'Setup' tab is selected, and the 'DHCP' sub-tab is active. The 'Internet Setup' section shows 'Automatic Configuration - DHCP' selected. The 'Network Setup' section shows the 'Router IP' as 192.168.0.1 and the 'Subnet Mask' as 255.255.255.0. The 'DHCP Server Settings' section shows the 'DHCP Server' is 'Enabled'. The 'Start IP Address' is 192.168.0.100, and the 'Maximum number of Users' is 50. The 'IP Address Range' is highlighted with a red box and shows '192.168.0. 100 - 149'. The 'Client Lease Time' is 0 minutes. The 'Static DNS' and 'WINS' sections are empty.

Admin

Physical Config **Desktop** Programming Attributes

Web Browser

URL: http://192.168.0.1 Go Stop

Wireless-N Broadband Router

Setup Wireless Security Access Restrictions Applications & Gaming Administration

Basic Setup DDNS MAC Address Clone Advanced Routing

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name: Domain Name: MTU: Size: 1500

Network Setup

Router IP: IP Address: 192 . 168 . 0 . 1 Subnet Mask: 255.255.255.0

DHCP Server Settings

DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.0. 100

Maximum number of Users: 50

IP Address Range: 192.168.0. 100 - 149

Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 0 . 0 . 0 . 0

Static DNS 2: 0 . 0 . 0 . 0

Static DNS 3: 0 . 0 . 0 . 0

WINS: 0 . 0 . 0 . 0

Help...

Step 4: Configure the Internet Port of WR.

In this step, **WR** is configured to route the packets from the wireless clients to internet. You will configure the **Internet** port on **WR** to connect to the internet.

- Under the **Internet Setup** at the top of the **Basic Setup** page, change the Internet IP address method from **Automatic Configuration – DHCP** to **Static IP**.
- Type the IP address to be assigned to the Internet interface as follows:

Internet IP Address: 209.165.200.225

Subnet Mask: 255.255.255.252

Default Gateway: 209.165.200.226

DNS Server: 209.165.201.1

The screenshot shows the Admin interface for a Wireless-N Broadband Router. The 'Internet Setup' tab is selected. The 'Internet Connection type' is set to 'Static IP'. The following fields are highlighted with a red box:

Internet IP Address:	209	165	200	225
Subnet Mask:	255	255	255	252
Default Gateway:	209	165	200	226
DNS 1:	209	165	201	1
DNS 2 (Optional):	0	0	0	0
DNS 3 (Optional):	0	0	0	0

Below the highlighted fields, there are fields for 'Host Name' and 'Optional Settings'.

- Scroll down the page and click **Save Settings**.

Note: If you get a **Request Timeout** message, close the Admin window and wait for the orange lights to turn into green triangles. Click the fast forward button to make this happen faster. Then reconnect to **WR** from **Admin's** browser using the process explained in Step 3.

- To verify connectivity, open a new web browser and navigate to **www.cisco.pka** server.

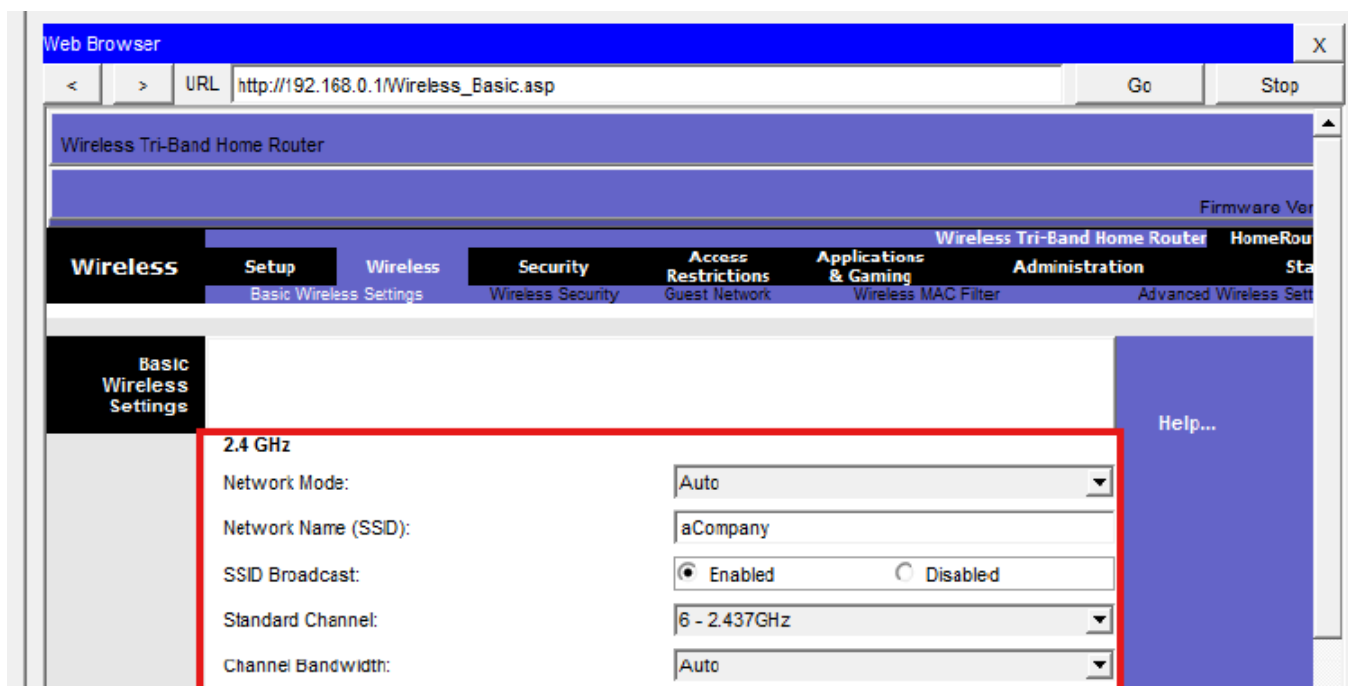
Note It may take a few seconds for the network to converge. Click **Fast Forward Time** or **Alt+D** to speed up the process.

Part 2: Configure the Wireless Settings

In this activity, you will only configure the wireless settings for 2.4 GHz.

Step 1: Configure the WR SSID.

- Navigate to the **WR** GUI interface at **192.168.0.1** in a web browser on **Admin**.
- Navigate to **Wireless > Basic Wireless Settings**.
- Change **Network Name (SSID)** to **aCompany** for only 2.4 GHz. Notice that SSIDs are case-sensitive.
- Change the **Standard Channel** to **6 - 2.437GHz**.
- For this activity, disable both 5 GHz frequencies. Leave the rest of the settings unchanged.
- Scroll to the bottom of the window and click **Save Settings**.

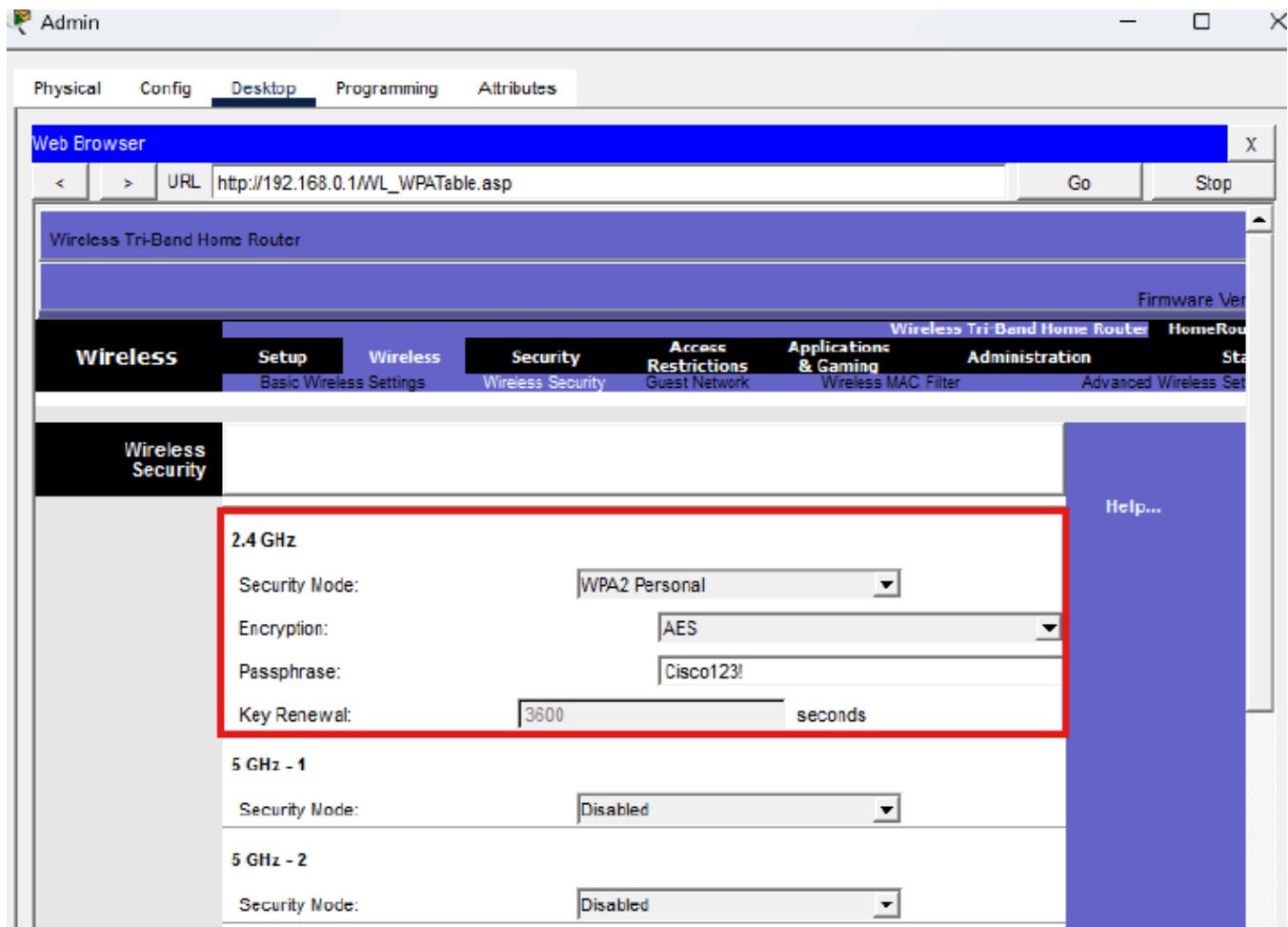


Step 2: Configure wireless security settings.

In this step, you configure the wireless security settings using WPA2 security mode with encryption and passphrase.

- Navigate to **Wireless > Wireless Security**.
- Under the 2.4 GHz heading, select **WPA2 Personal** for the Security Mode.
- For the Encryption field, keep the default **AES** setting.
- In the Passphrase field, enter **Cisco123!** as the passphrase.
- Click **Save Settings**.

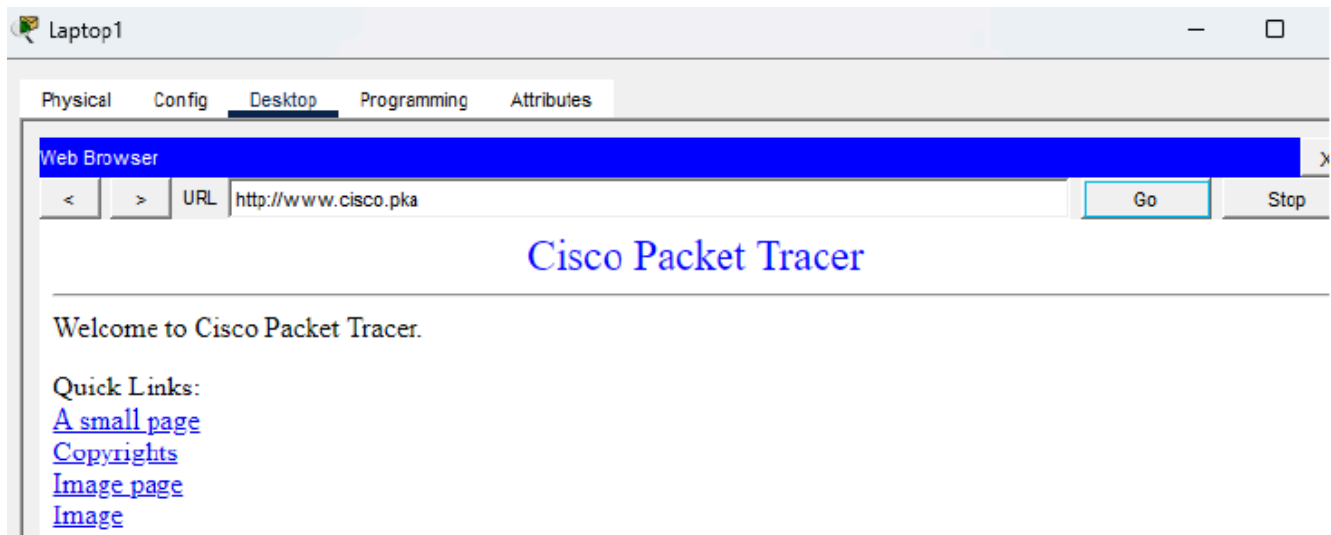
- f. Verify that the settings in the **Basic Wireless Settings** and **Wireless Security** pages are correct and saved.

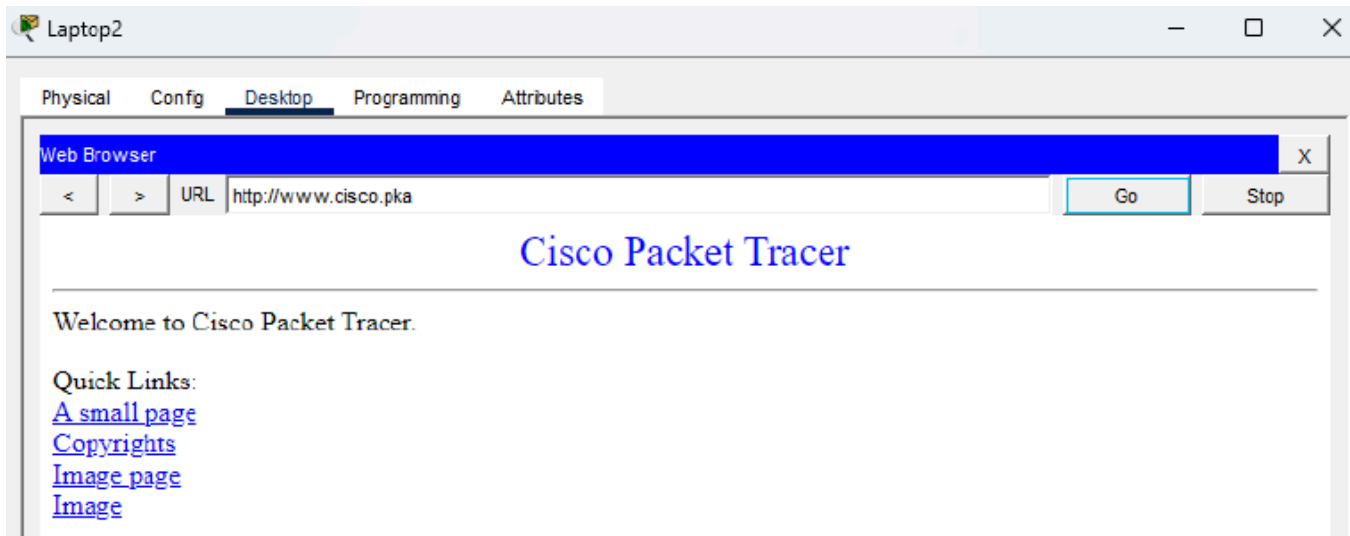
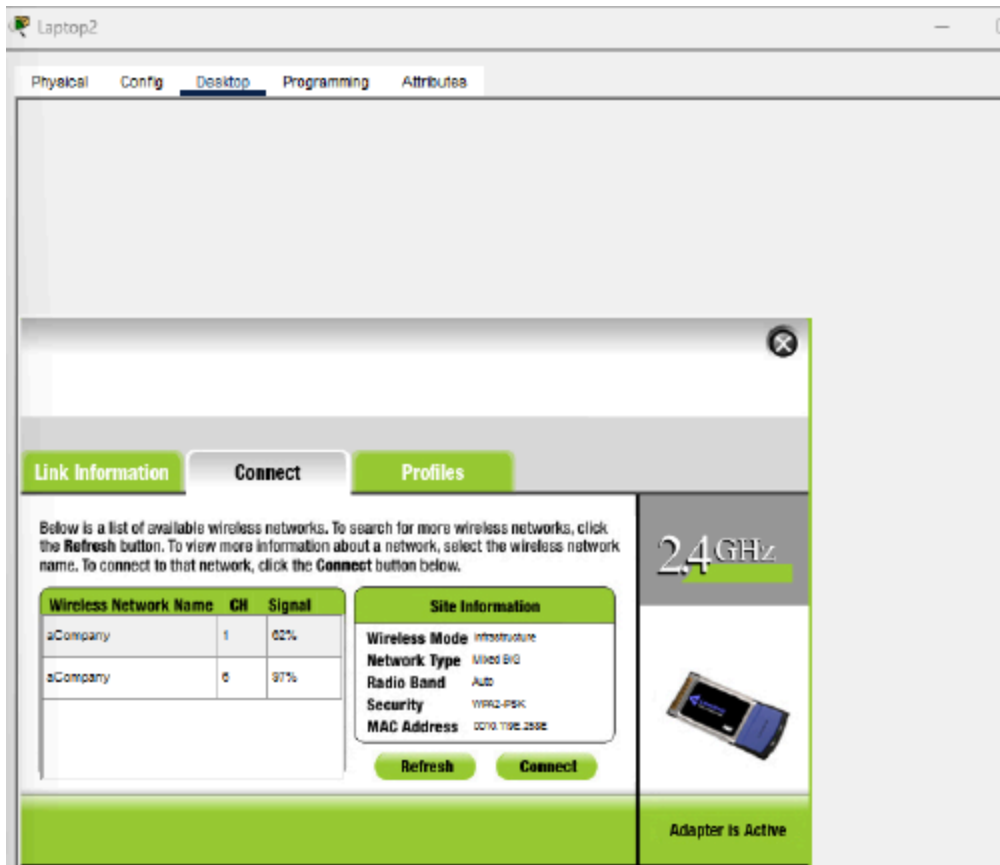


Step 3: Connect the Wireless Clients.

- Open **Laptop1**. Select **Desktop** tab. Click **PC Wireless**.
- Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **aCompany**.
- Enter the passphrase configured in the previous step. Enter **Cisco123!** in the pre-shared key field and click **Connect**. Close the PC Wireless window.
- Open a web browser and verify that you can navigate to **www.cisco.pka** server.
- Repeat the above steps to connect **Laptop2** to the wireless network.

Packet Tracer - Configure a Wireless Network



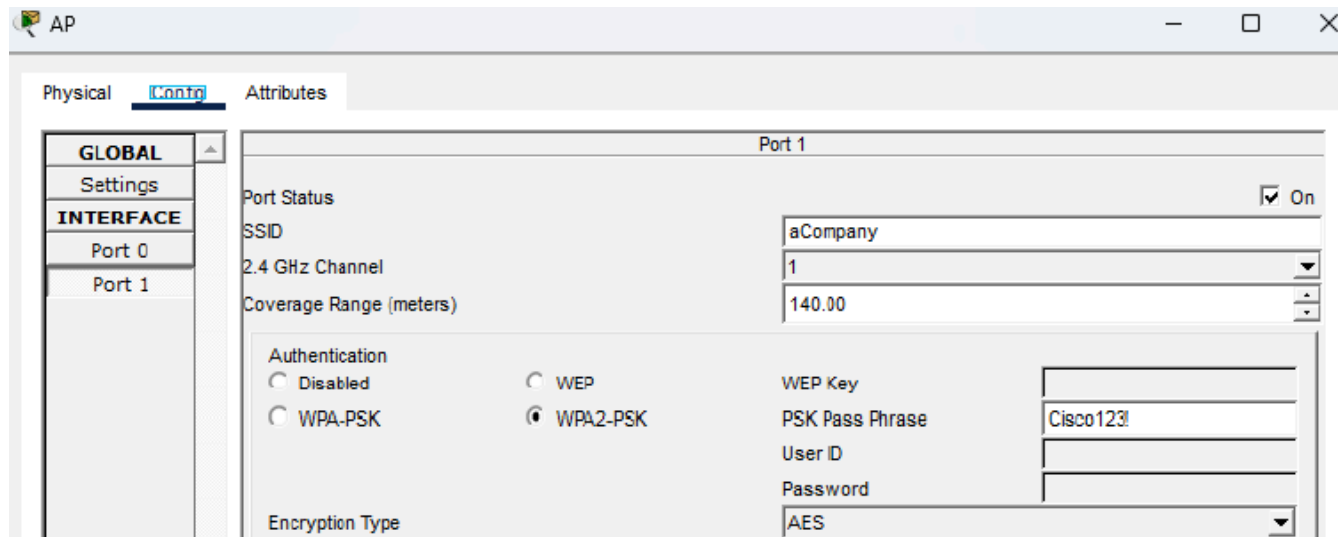


Part 3: Connect Wireless Clients to an Access Point

An access point (AP) is a device that extends the wireless local area network. An access point is connected to a wired router using an Ethernet cable to project the signal to a desired location.

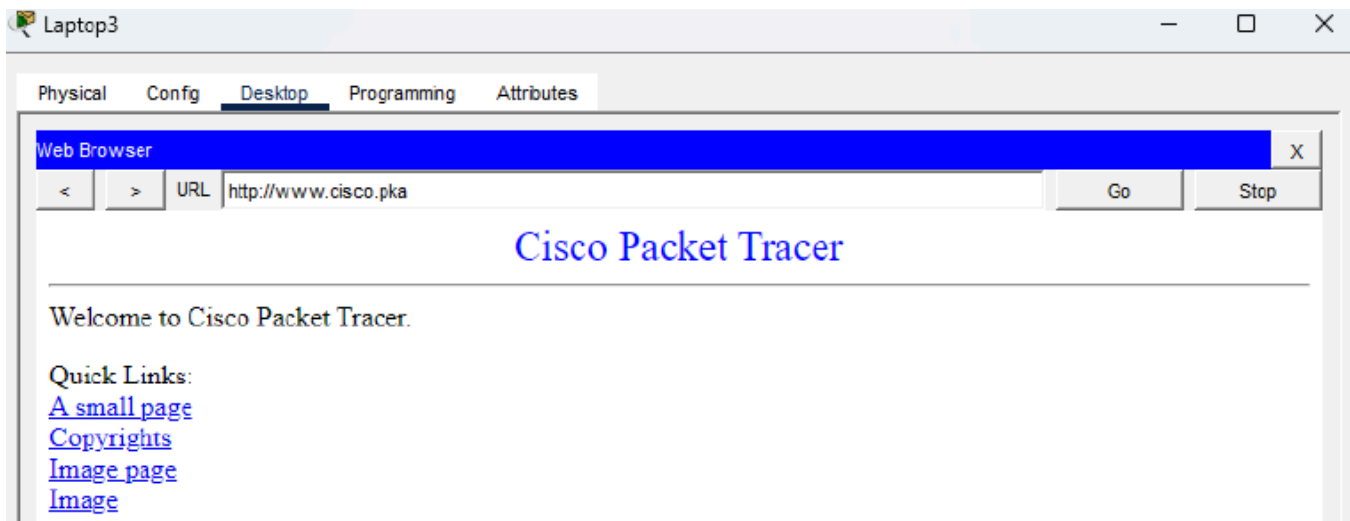
Step 1: Configure the Access Point.

- Connect **Port 0** of **AP** to an available Ethernet port of **WR** using a straight-through Ethernet cable.
- Click **AP**. Select the **Config** tab.
- Under the **INTERFACE** heading, select **Port 1**.
- In the **SSID** field, enter **aCompany**.
- Select **WPA2-PSK**. Enter the passphrase **Cisco123!** in the **Pass Phrase** field.
- Keep **AES** as the default **Encryption Type**.



Step 2: Connect the Wireless Clients.

- Open **Laptop3**. Select **Desktop** tab. Click **PC Wireless**.
- Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **aCompany** with the stronger signal (Channel 1) and click **Connect**.
- Open a web browser and verify that you can navigate to **www.cisco.pka** server.

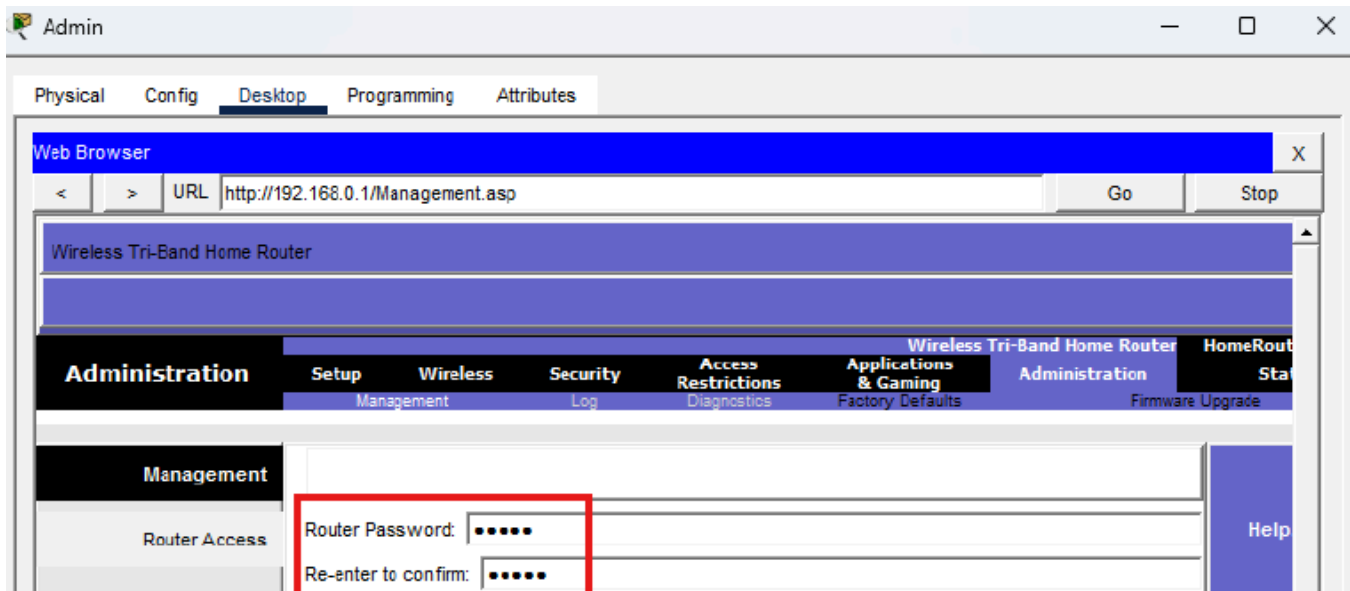


Part 4: Other Administrative Tasks

Step 1: Change the WR Access Password.

- On **Admin**, navigate to WR GUI interface at **192.168.0.1**.
- Navigate to **Administration > Management** and change the current **Router Password** to **cisco**.
- Scroll to the bottom of the window and click **Save Settings**.

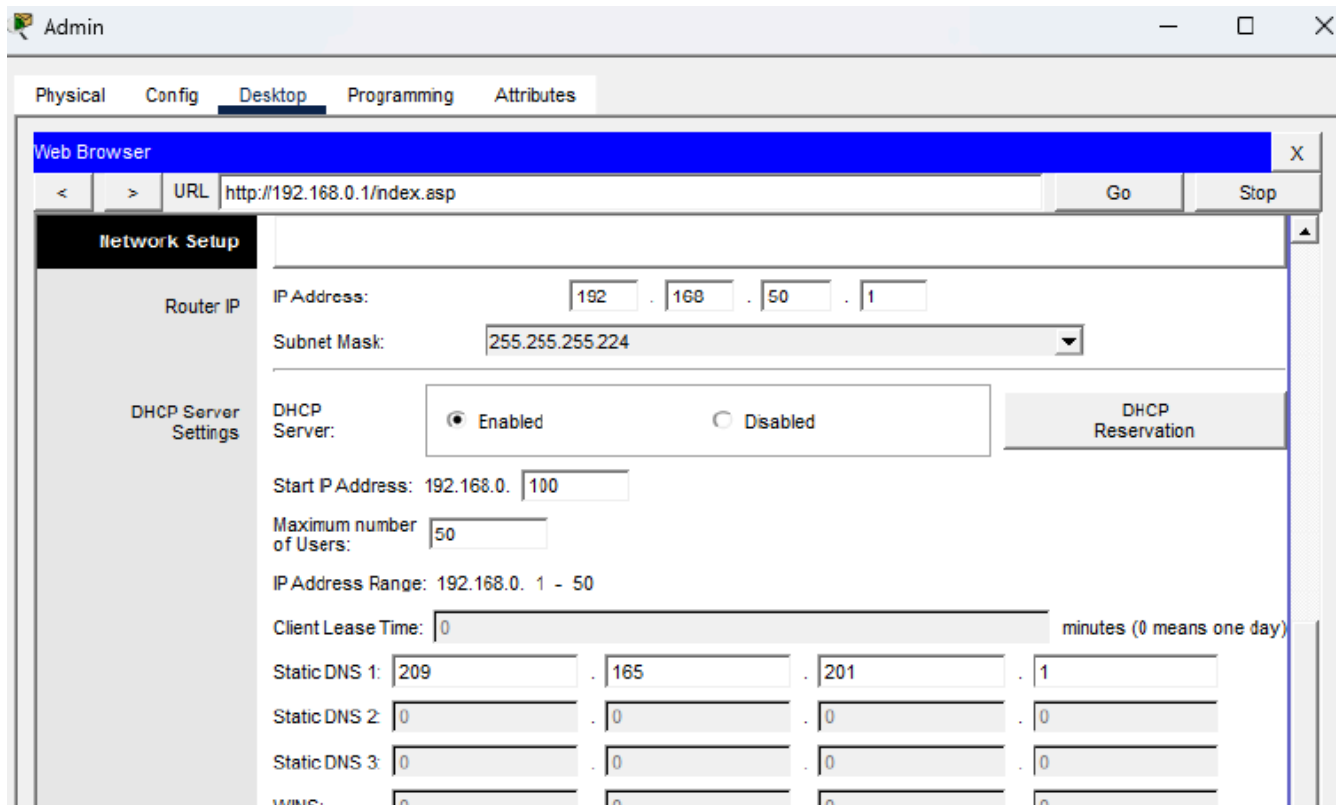
- d. Use the username **admin** and the new password **cisco** when prompted to log in to the wireless router. Click **OK** to continue.
- e. Click **Continue** and move on to the next step.



Step 2: Change the DHCP address range in WR.

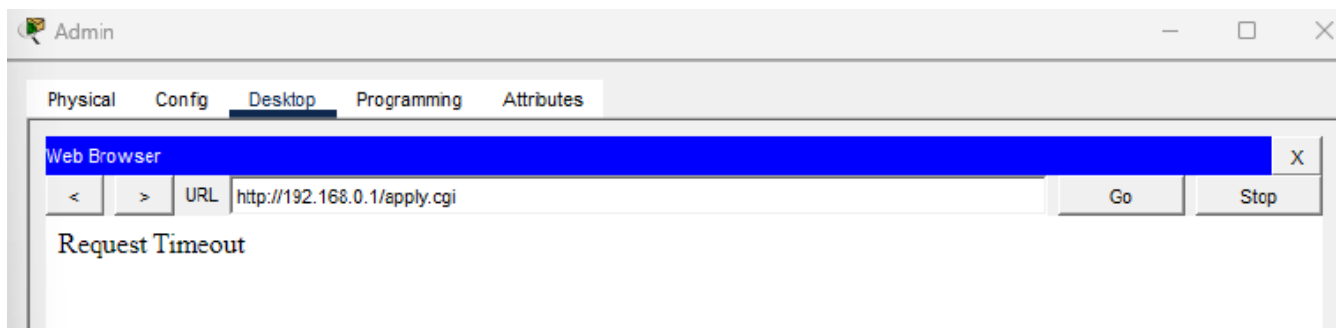
In this step, you will change the internal network address from 192.168.0.0/24 to 192.168.50.0/24. When the LAN network address changes, the IP addresses on the devices in the LAN and WLAN must be renewed to receive new IP addresses before the lease is timed out.

- a. Navigate to **Setup > Basic Setup**.
- b. Scroll down the page to **Network Setup**.
- c. The IP address assigned to **Router IP** is 192.168.0.1. Change it to 192.168.50.1. Verify that IP address still start at .100, and there are 50 available IP addresses in the DHCP pool.
- d. Add **209.165.201.1** as the DNS server with the DHCP settings.
- e. Scroll to the bottom of the window and click **Save Settings**.



- f. Note that the DHCP range of addresses has been automatically updated to reflect the interface IP address change. The Web Browser will display a **Request Timeout** after a short time. Why?

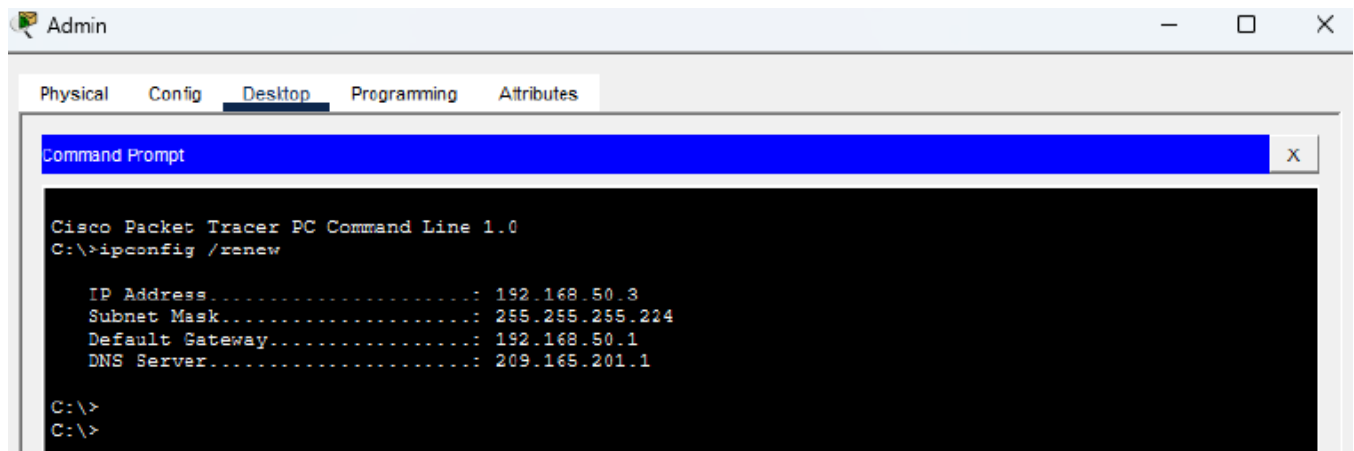
Modifying the router's internal gateway address breaks the session because it places the router and the Admin PC on two separate subnets. The router immediately applies the new network settings, but the PC continues to use its old IP configuration. Without a refreshed IP address that matches the router's new subnet, the PC cannot establish contact.



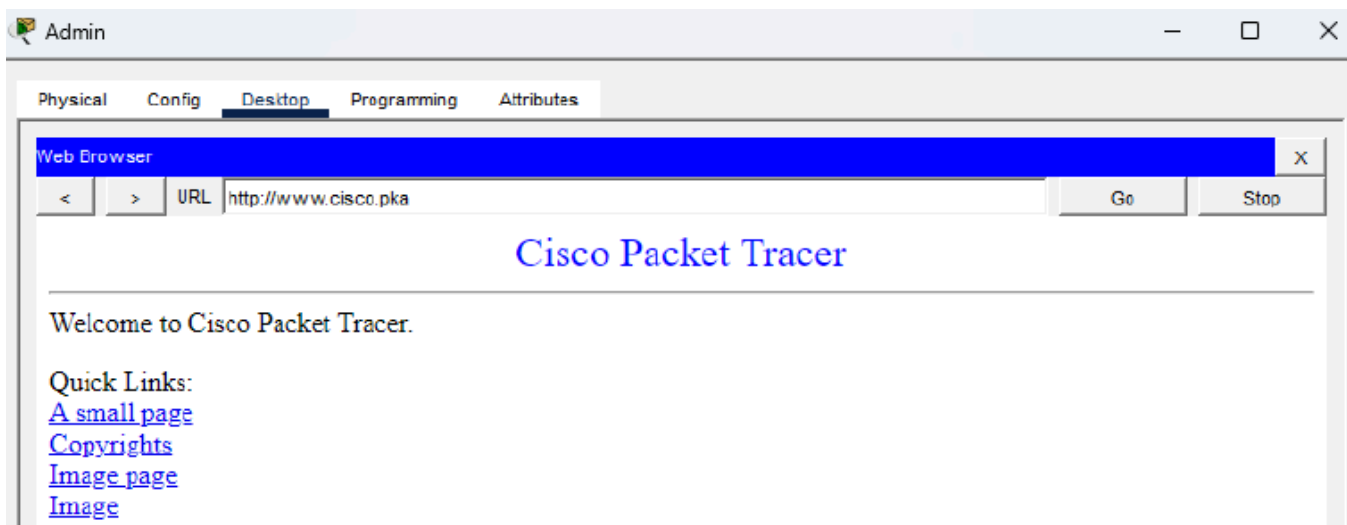
- g. Close the **Admin** web browser.
 h. In **Admin Desktop** tab, click **Command Prompt**.
 i. Type **ipconfig /renew** to force **Admin** re-acquire its IP information via DHCP.

What is the new IP address information for **Admin**?

192.168.50.3

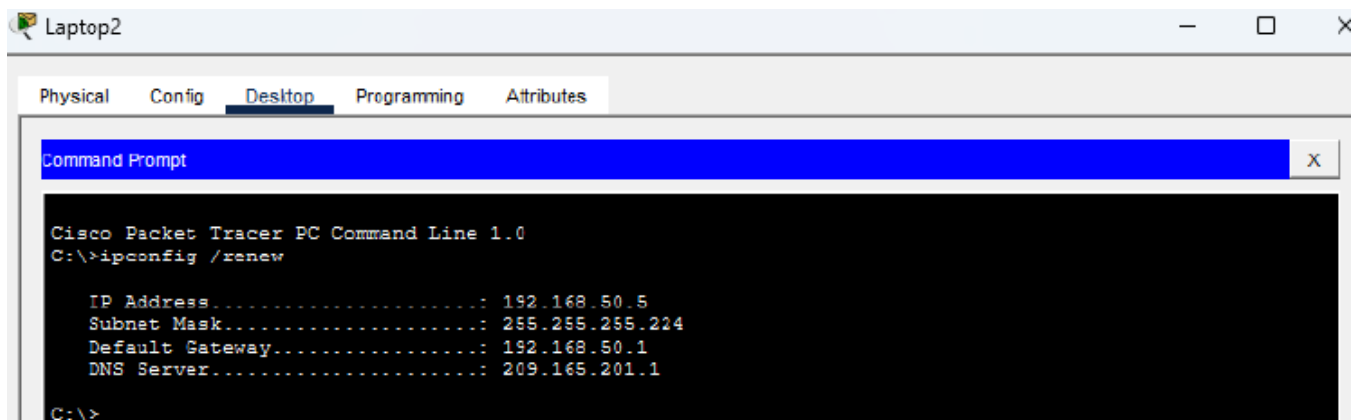
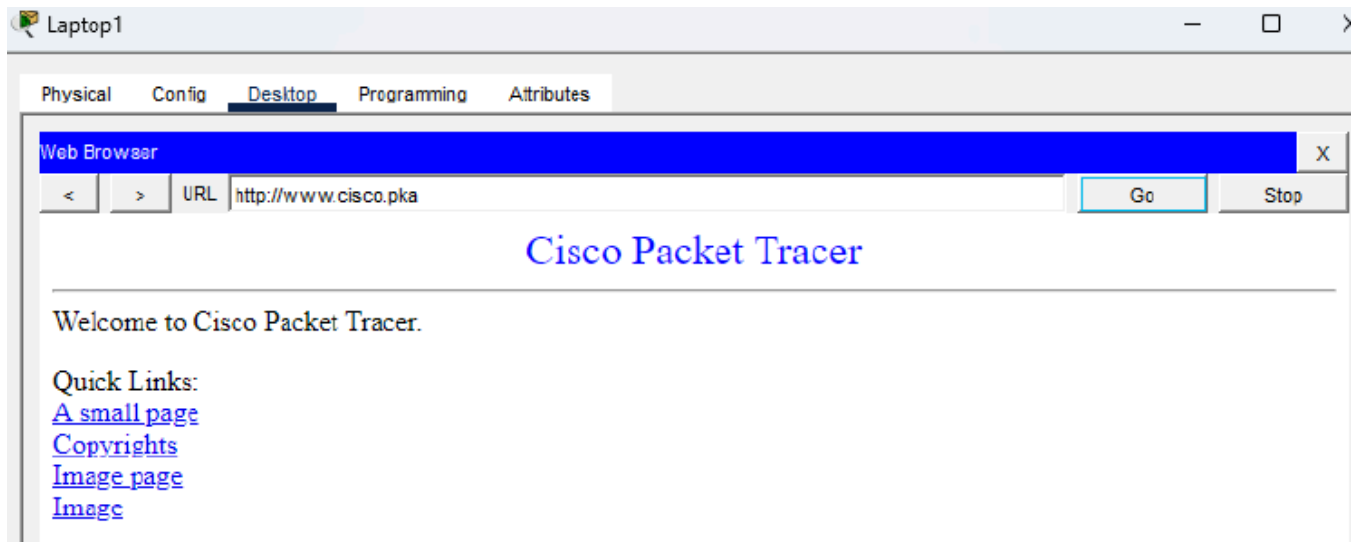
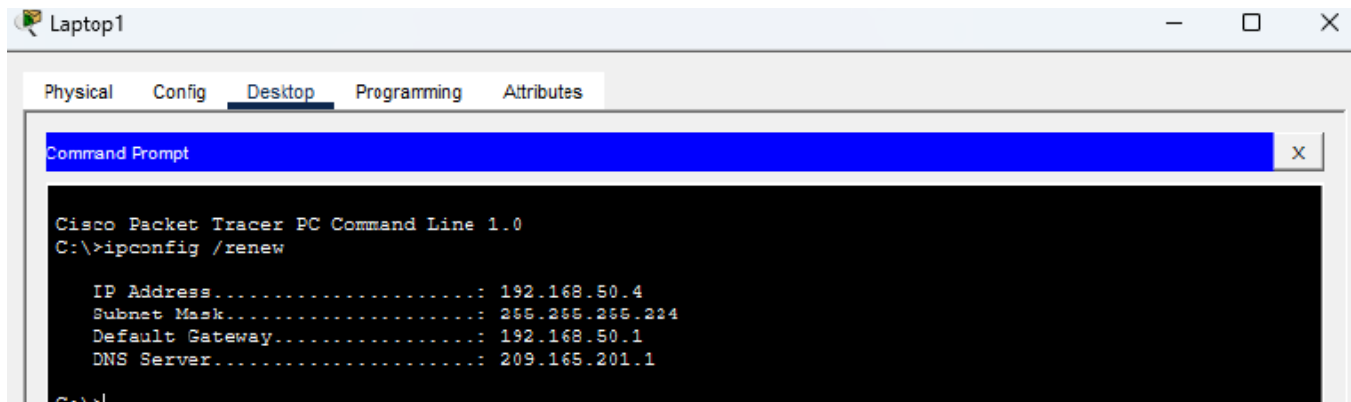


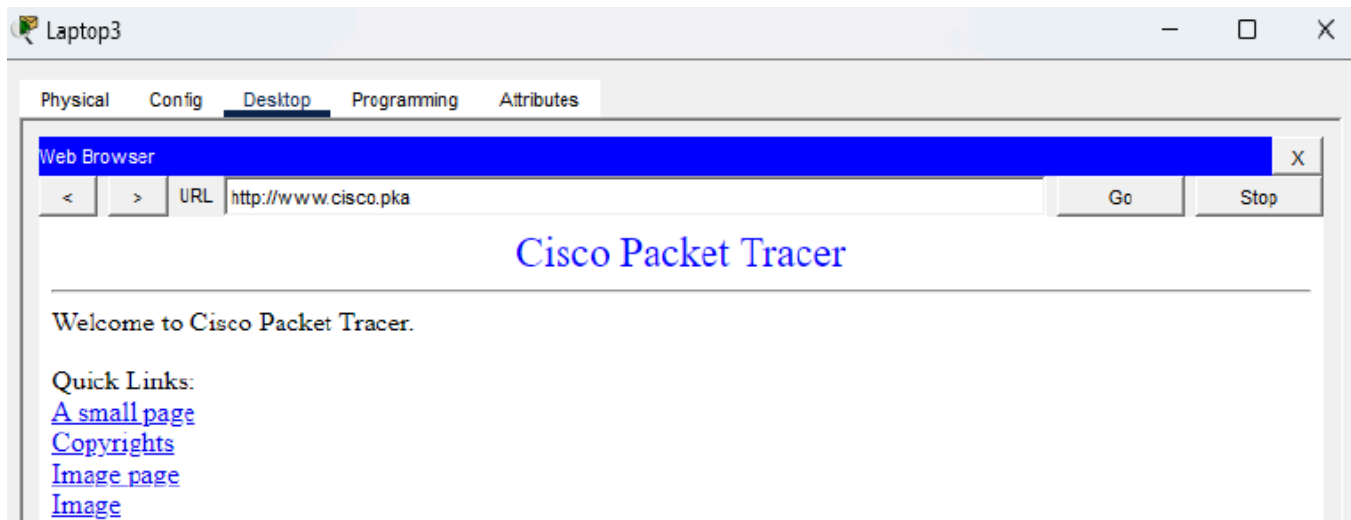
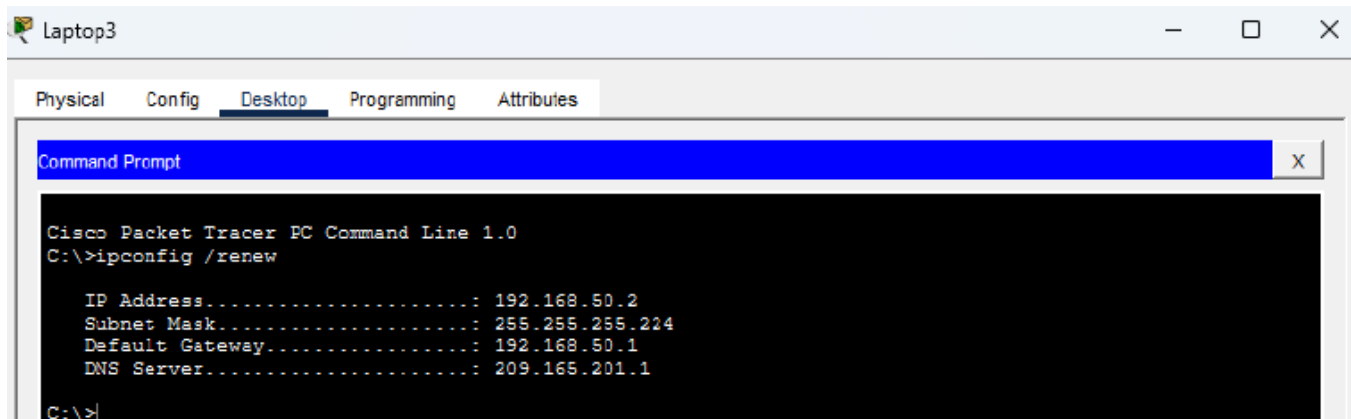
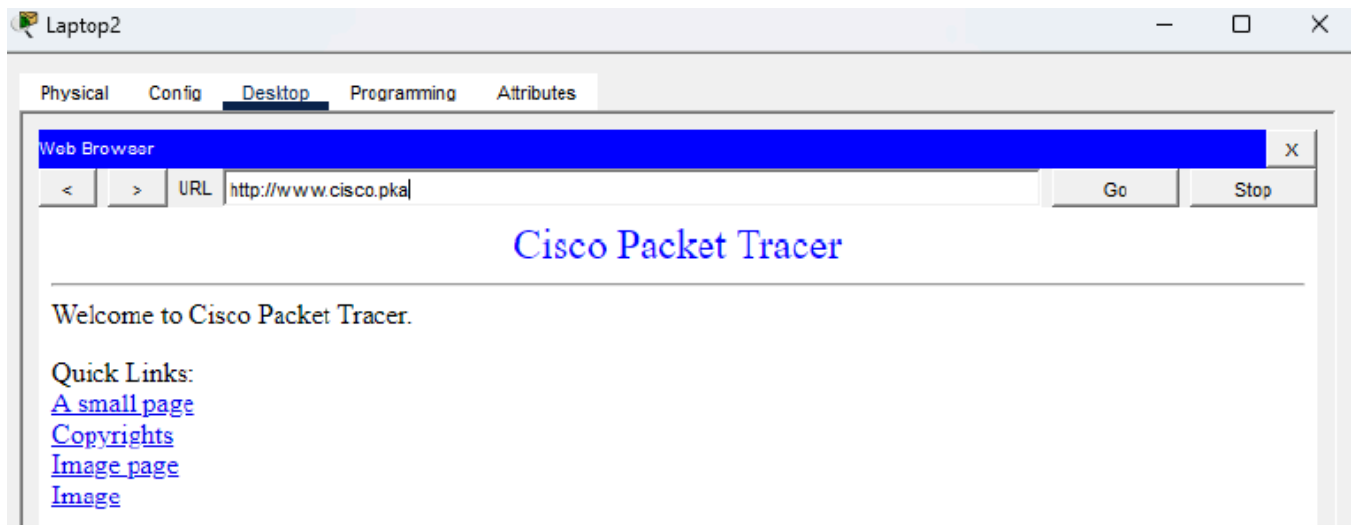
j. Verify that you can still navigate to **www.cisco.pka** server.



k. Renew the IP address on other laptops to verify that you can still navigate to **www.cisco.pka** server.

Packet Tracer - Configure a Wireless Network





- I. Notice that **Laptop1** connected to the **AP** instead of **WR**.

Why?

Laptop1 connects to the AP because both the wireless router (WR) and the access point (AP) share the exact same SSID ("aCompany") and WPA2 passphrase. This setup creates an Extended Service Set (ESS). When a wireless device detects multiple transmitters for the same network, it automatically pairs with the one offering the strongest signal. In this topology, the AP provides a higher Received Signal Strength Indicator (RSSI) to Laptop1 than the WR, prompting the laptop to choose the AP for a better connection.